

positive and negative terms; otherwise, no pinched would occur and the integration contour can be deformed, avoiding the singularity of the integrand.

This Landau-equation interpretation suggests a search algorithm to identify graphs that potentially contain hidden regions for given kinematics [20]. The idea is to examine the existence of the super-leading $\mathcal{F}$ terms, whose sum $\mathcal{F}_{\text{SL}}^{(R)}(\mathbf{x}; \mathbf{s})$, according to the above, must satisfy:

$$\frac{\partial \mathcal{F}_{\text{SL}}^{(R)}}{\partial x_e} \text{ contains both signs if it is nonzero.}$$

For a given graph, one can focus on the first derivatives of the $\mathcal{F}$ polynomial. If $\partial \mathcal{F} / \partial x_{e_0}$ is positive or negative definite for a given edge e_0 , then the parameter x_{e_0} cannot be contained in $\mathcal{F}_{\text{SL}}^{(R)}(\mathbf{x}; \mathbf{s})$, and we can simply focus on the first derivatives of the remaining terms, $\mathcal{F}|_{x_{e_0}=0}$ recursively. If we end up with a nonzero $\mathcal{F}_{\text{SL}}^{(R)}(\mathbf{x}; \mathbf{s})$, then the graph potentially features a hidden region R ; otherwise, all its regions are facet regions.

For each graph with a nonzero $\mathcal{F}_{\text{SL}}^{(R)}(\mathbf{x}; \mathbf{s})$, to explicitly reveal its hidden regions we dissect the original polytope $\Delta(\mathcal{P})$ into several sub-polytopes based on the structure of the $\mathcal{F}_{\text{SL}}^{(R)}$ polynomial. The dissection separates the domain of integration on the hypersurfaces where the derivatives of $\mathcal{F}_{\text{SL}}^{(R)}$ vanish. Each sub-polytope is thus free said cancellations and its facets determine the corresponding asymptotic expansion. A convenient way to implement that is to map these sub-polytope integrals back to the usual integration domain through a suitable change of variables, such that the original pinch singularities map to the integration endpoints ($x_e = 0$ and ∞) in the new parameter space, thereby converting the hidden regions that were located in the interior of the original $\Delta(\mathcal{P})$ into facet regions of the resulting sub-polytopes in the new parameter space.

Below we shall apply the general analyses above to the $2 \rightarrow 3$ spacelike-collinear limit of the five-point scattering amplitude at two loops.

5.2 Identifying hidden regions

Consider a Lee–Pomeransky representation $\mathcal{P}(\mathbf{x}, \mathbf{s}) = \mathcal{U}(\mathbf{x}) + \mathcal{F}(\mathbf{x}; \mathbf{s})$, where $\delta \in \mathbf{s}$ denotes the expansion parameter. Let $\mathcal{K}$ denote the kinematic domain relevant for the asymptotic limit under consideration. Below we present an algorithm whose goal is to determine whether $\mathcal{F}$ contains a hidden region in the kinematic domain $\mathcal{K}$ and then fully characterize that region. The key condition is that $\mathcal{F}$ would contain a subset of monomials that are superleading in δ , and furnish a cancellation sector, such that their total contribution is of the same order as other (leading) terms in $\mathcal{P}$. The challenge is that the superleading nature of these terms would only become apparent after a suitable rescaling of the Lee–Pomeransky parameters is introduced. The algorithm is designed to find that scaling law.

Before applying any scaling, the Taylor expansion of the second Symanzik polynomial may be written as³

$$\mathcal{F}(\mathbf{x}, \mathbf{s}) = \mathcal{F}_0(\mathbf{x}, \mathbf{s}) + \sum_{n=1}^{\infty} \delta^n \mathcal{F}_n(\mathbf{x}, \mathbf{s}) \equiv \mathcal{F}_0(\mathbf{x}, \mathbf{s}) + \mathcal{F}_\delta(\mathbf{x}, \mathbf{s}), \quad (5.7)$$

³For simplicity of notation, in what follows we will often omit the arguments in $\mathcal{F}_0$ and $\mathcal{F}_\delta$.

where δ dependence arises from the external kinematics only – we have not yet scaled the Lee-Pomeransky parameters. We will assume here that the superleading monomials (i.e. the cancellation structure) are fully captured by the leading order polynomial $\mathcal{F}_0$. The generalization to the case that $\mathcal{F}_\delta$ is relevant is straightforward, but we have no example where it is needed.

Cancellation factors We first form the set of (non-monomial) factors appearing either in the derivatives of the strict leading polynomial or in coefficients of independent kinematic structures appearing in those derivatives.

$$\mathcal{DF}_0 = \left\{ f(\mathbf{x}, \mathbf{s}) : f \text{ is a non-monomial factor appearing in } \frac{\partial \mathcal{F}_0}{\partial x_e} \text{ for some } e \right\}. \quad (5.8)$$

From this set we retain only mixed-sign polynomials factors

$$f(\mathbf{x}, \mathbf{s}) = \sum_i c_i(\mathbf{s}) \mathbf{x}^{\mathbf{r}_i} \quad (5.9)$$

which consist of *at least* one positive and one negative monomials for $x_i > 0$, given the kinematic domain $\mathcal{K}$ in which the expansion is done. Thanks to the mixed signs, the factors $f(\mathbf{x}, \mathbf{s})$ can participate in an admissible cancellation locus. We denote the resulting set by

$$\mathcal{C}(\mathcal{F}_0; \mathcal{K}) \subset \mathcal{DF}_0. \quad (5.10)$$

A subset of $\mathcal{C}(\mathcal{F}_0; \mathcal{K})$

$$S = \{f_1, \dots, f_r\} \subseteq \mathcal{C}(\mathcal{F}_0; \mathcal{K}) \quad (5.11)$$

is called *admissible* if

$$\exists (\mathbf{x}, \mathbf{s}) \in \mathbb{R}_{>0}^n \times \mathcal{K} \quad \text{such that} \quad f_1(\mathbf{x}, \mathbf{s}) = \dots = f_r(\mathbf{x}, \mathbf{s}) = 0. \quad (5.12)$$

Note that the solution must simultaneously admit the requirement that $\mathbf{s}$ is in the relevant kinematic domain and that x_e are all positive, which is the standard positivity condition associated with first-sheet Landau singularities. The set $\mathcal{C}(\mathcal{F}_0; \mathcal{K})$ contains all admissible candidate factors identified from the derivative analysis. At this stage no preferred subset is singled out. Determining which subset participates in a hidden region is part of the obstruction analysis described below.

A necessary condition for a hidden region is the existence of an admissible subset $S = \{f_1, \dots, f_r\}$ with $r \geq 2$. Indeed, a single factor merely defines a hypersurface, whereas a hidden region requires a nontrivial cancellation structure arising from the simultaneous vanishing of multiple factors.

Identification of the obstruction As discussed above, superleading terms in a hidden region form a factorized structure, denoted $\mathcal{F}_{\text{SL}}$, which admits internal cancellation when the scaling law of the Lee-Pomeransky parameters is accounted for. A key challenge in

identifying the hidden region is that generally $\mathcal{F}_0$ contains, alongside the superleading terms, additional terms, which we shall refer to as *obstruction*. The general expected structure is

$$\mathcal{F}_0 = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obst}}, \quad (5.13)$$

where $\mathcal{F}_{\text{Obst}}$ would ultimately be suppressed compared to the superleading monomials when accounting for the hidden-region scaling law. We stress that at this point neither of the terms in Eq. (5.13) is known, nor is said scaling law. The purpose of the obstruction analysis is to simultaneously determine $\mathcal{F}_{\text{Obst}}$ and $\mathcal{F}_{\text{SL}}$. Let

$$\mathcal{F}_0 = \sum_n m_n(\mathbf{x}, \mathbf{s}) \quad (5.14)$$

be the expansion of $\mathcal{F}_0$ into monomials.

Given an admissible subset

$$S = \{f_1, \dots, f_r\} \subseteq \mathcal{C}(\mathcal{F}_0; \mathcal{K}), \quad (5.15)$$

we search for a subset of original monomials $\{m_i(\mathbf{x}, \mathbf{s})\}$, whose sum defines $\mathcal{F}_{\text{Obst}}$, such that when removed from $\mathcal{F}_0$, then the remaining terms can be organized as follows

$$\mathcal{F}_{\text{SL}} = \sum_k M_k(\mathbf{x}, \mathbf{s}) g_k(\mathbf{x}, \mathbf{s}) \quad \text{with} \quad g_k(\mathbf{x}, \mathbf{s}) = \prod_{f_i \in S_k} f_i(\mathbf{x}, \mathbf{s}) \quad (5.16)$$

for some polynomials $M_k(\mathbf{x}, \mathbf{s})$. Each $g_k(\mathbf{x}, \mathbf{s})$ admits a factorised structure with *at least two* factors $f_i, f_j \in S_k$. Equivalently, the defining property of $\mathcal{F}_{\text{Obst}}$ is that upon subtracting it from $\mathcal{F}_0$, the remaining polynomial $\mathcal{F}_{\text{SL}}$ belongs to the ideal generated by G , namely

$$\mathcal{F}_0 - \mathcal{F}_{\text{Obst}} = \mathcal{F}_{\text{SL}} \in I_{\text{canc}}(G) \equiv \langle G \rangle = \langle g_1, g_2, \dots, g_{n_G} \rangle, \quad (5.17)$$

where $G = \{g_1, g_2, \dots, g_{n_G}\}$ is a generator set associated with the admissible subset S . Each generator g_k takes the form of a product of the factors f_i , for example we may have $G = \{f_1 f_2\}$ for $n_G = 1$ and $G = \{f_1 f_2, f_3 f_4\}$ for $n_G = 2$. An additional important requirement is that the generators g_k contribute at the same scaling order in the hidden region, so that all terms appearing in $\mathcal{F}_{\text{SL}}$ in Eq. (5.16) can simultaneously participate in the leading cancellation structure. If several decompositions satisfy the ideal-membership condition, the one with the minimal obstruction is selected.

Determination of the hidden region scaling Once a hidden-region obstruction has been identified, the associated scaling vector is determined from the monomial structure of the Lee–Pomeransky polynomial $\mathcal{P}(\mathbf{x}, \mathbf{s})$ and the requirement that it yields a scalefull integral. Let us now restore the complete $\mathcal{F}$ polynomial including the δ -suppressed terms of (5.7) we left behind. The complete Lee–Pomeransky polynomial $\mathcal{P}$ is

$$\mathcal{P} = \mathcal{U} + \mathcal{F}, \quad \text{with} \quad \mathcal{F} = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{Obs}} + \mathcal{F}_{\delta}, \quad (5.18)$$

where $\mathcal{F}_{\text{SL}}$ denotes the collection of superleading monomials forming the cancellation and $\mathcal{F}_{\text{Obs}}$ contains the remaining contributions to $\mathcal{F}_0$. Assigning scaling weights $\{v_e\}$,

$$x_e \rightarrow x_e \delta^{v_e}, \quad (5.19)$$

to the Lee-Pomeransky parameters, each monomial (5.3)

$$m_i = \bar{c}_i(\mathbf{s}) \delta^{a_i} x_1^{r_{i,1}} \cdots x_N^{r_{i,N}}, \quad \text{for } c_i(\mathbf{s}) \sim \mathcal{O}(\delta^{a_i}), \quad (5.20)$$

acquires a weight

$$w(m_i) = a_i + \sum_e r_{i,e} v_e = a_i + \mathbf{r}_i \cdot \mathbf{v}. \quad (5.21)$$

To have a physical intuition of the scaling law it is useful to keep in mind that the scaling of the virtuality of a given propagator e is inversely proportional to the scaling of the corresponding Lee-Pomeransky parameter [21]. In particular, infrared regions are characterized by negative exponents v_e , so the dominant monomials are those carrying the minimal, or most negative weights.

In order to determine the scaling vector $\mathbf{v}$ of the hidden region we impose several requirements that stem from the cancellation structure of $\mathcal{F}_{\text{SL}}$ and the requirement that the corresponding region integral would be scaleful. That integral is defined by (5.1) with the Lee-Pomeransky polynomial considered at leading order in δ after applying the $\mathbf{v}$ scaling.

The first requirement is that all monomials belonging to $\mathcal{F}_{\text{SL}}$ scale identically. Since these monomials cancel among themselves, any difference in their weights would prevent the cancellation from taking place in the asymptotic limit. The scaling vector must therefore satisfy

$$W_{\text{SL}} \equiv w(m_1) = w(m_2) = \cdots = w(m_k), \quad \forall m_i \in \mathcal{F}_{\text{SL}}. \quad (5.22)$$

These conditions must strictly hold for all the monomials in $\mathcal{F}_{\text{SL}}$. Hence, they form a system of linear equations for the scaling exponents $\{v_e\}$, the components of the scaling vector $\mathbf{v}$.

We further define the hidden-region weight, namely the way the $\mathcal{P}$ ultimately scales with δ , as

$$W_{\text{HR}} \equiv \min\{w(m_i)\} \quad \forall m_i \in \mathcal{P} \setminus \mathcal{F}_{\text{SL}}. \quad (5.23)$$

We recall that for infrared regions the weights are expected to be negative, so the minimization picks the largest monomials not in $\mathcal{F}_{\text{SL}}$ i.e., those which are the most singular in the limit $\delta \rightarrow 0$.

The cancellation in $\mathcal{F}_{\text{SL}}$ brings it from being superleading to being leading, i.e. being comparable to other leading terms inside in $\mathcal{P}$ in Eq. (5.18). After the cancellation among the monomials of $\mathcal{F}_{\text{SL}}$, the surviving contribution is required to effectively behave as $\delta^{W_{\text{HR}}}$. If the cancellation were to suppress $\mathcal{F}_{\text{SL}}$ beyond this order, the resulting leading polynomial would not contain the hidden-region contribution, and there would not be a hidden region. The cancellation requirement is $W_{\text{HR}} > W_{\text{SL}}$. Thus, by definition for any monomial $m_i \in \mathcal{P}$ which is not part of the superleading $\mathcal{F}_{\text{SL}}$, i.e. in $\mathcal{F}_{\text{Obs}}$ in $\mathcal{F}_{\text{Obs}}$ and in U , we have

$$w(m_i) \geq W_{\text{HR}} > W_{\text{SL}}, \quad \forall m_i \in \mathcal{P} \setminus \mathcal{F}_{\text{SL}} \quad (5.24)$$

This strict inequality is an additional requirement for any solution for the scaling vector $\mathbf{v}$, beyond the linear equations of Eq. (5.22).

Finally, for the hidden region integral to be scalefull (that is not vanish) it is essential that it would depend on all active Lee-Pomeransky parameters (see the discussion below regarding boundary hidden regions). Equivalently, every active (i.e. non-vanishing) parameter must appear in at least one monomial m_i contributing to the leading asymptotic support (after the cancellation of $\mathcal{F}_{\text{SL}}$). The requirement is

$$\forall x_e \neq 0, \quad \exists m_i \in \mathcal{P} \quad \text{such that} \quad x_e \in m_i \quad \text{and} \quad w(m_i) \in \{W_{\text{HR}}, W_{\text{SL}}\}. \quad (5.25)$$

This final condition on the hidden region scaling vector $\mathbf{v}$ excludes residual scaleless integrations. We expect that this implies in particular that the leading monomials in U would have the scaling W_{HR} .

The scaling vector is therefore determined by a combination of linear equality constraints originating from the superleading cancellation and inequality constraints expressing the hierarchy and non-scalelessness conditions. In all examples considered so far, these constraints are sufficiently restrictive to determine a unique admissible scaling vector.

Boundary hidden regions The preceding discussion assumes that the hidden region is supported in the interior of the Lee–Pomeransky integration domain. We shall refer to it as an *interior hidden region*. Typically, this is the situation in the simplest topology for which a hidden region occurs for a given kinematic setup, but for graphs with additional propagators, the relevant cancellation structure may instead appear only on a boundary stratum, where one or more parameters are set to zero. We shall refer to these as *boundary hidden regions*. This is the familiar situation from Landau analysis often classified as subleading Landau singularities.

To identify boundary hidden regions we therefore repeat the obstruction analysis on strata of the form

$$x_{e_1} = \dots = x_{e_k} = 0, \quad (5.26)$$

where $e_1, e_2 \dots e_k$ are a subset of edges in the graph. On such a stratum we restrict the strict leading polynomial,

$$\mathcal{F}_0 \longrightarrow \mathcal{F}_0|_{x_{e_1}=\dots=x_{e_k}=0}, \quad (5.27)$$

remove the variables which are set to zero from the positivity constraints, and repeat the construction of admissible factors, obstructions and superleading terms.

Overall, the algorithm discussed thus far allows us to distinguish four scenarios:

- an interior hidden region, where the hidden region is identified with no variables set to zero, while setting to zero any of the variables excludes the cancellation structure;
- a boundary hidden region, which is identified according to the conditions discussed above only after setting one or more Lee–Pomeransky parameters to zero;
- a hidden region which is present in the interior as well as on the boundary;

- no detected hidden region, where there is no admissible cancellation sector under the assumptions we have made.

In either of the first three cases the hidden region would affect the asymptotic expansion of integral. The boundary scan is therefore an essential part of the algorithm.

A posteriori consistency check As emphasised at the outset, our construction starts from the strict leading polynomial $\mathcal{F}_0$. It therefore assumes that the terms relevant for the superleading cancellation sector are already present at leading order in the fixed-parameter expansion in δ .

This assumption can be checked a posteriori using the full polynomial (5.7). Consider a higher-order monomial $m_i \in \mathcal{F}_\delta$, where, prior to scaling the Lee-Pomeransky parameters scales as δ^{a_i} for some $a_i \geq 1$. The total weight of this monomial in the hidden region is $w(m_i) = a_i + \mathbf{r}_i \cdot \mathbf{v}$. Our cancellation-sector analysis based on $\mathcal{F}_0$ is self-consistent provided that no such δ -suppressed term is promoted into the superleading polynomial $\mathcal{F}_{\text{SL}}$ through the scaling $\mathbf{v}$. In other words, after the scaling vector has been determined, one must verify that there is no term $m_i \in \mathcal{F}_\delta$ (i.e. a term having $a_i \geq 1$) satisfying

$$w(m_i) = a_i + \mathbf{r}_i \cdot \mathbf{v} \leq W_{\text{SL}}. \quad (5.28)$$

If such a term exists, then the proposed superleading polynomial is incomplete: the promoted terms must be included in $\mathcal{F}_{\text{SL}}$ and the obstruction analysis must be repeated. If no such term exists, the hidden region found from $\mathcal{F}_0$ is stable under the full δ -dependent polynomial. Presently, all known examples of hidden regions are compatible with this criterion, in other words the strict limit, $\mathcal{F}_0$, is sufficient to algorithmically determine the hidden region.

5.3 Revealing hidden regions via polytope dissection

We now apply the search algorithm explained above to the case of five massless particle scattering at two loops in order to exclude graphs where no hidden regions can arise, and thus single out those with non-vanishing $\mathcal{F}_{\text{SL}}^{(R)}$. Most graphs, including all planar graphs, belong to the former category. For example, the $\mathcal{F}$ polynomial of the one-loop graph in (see figure 3a) is:

$$\mathcal{F}(\mathbf{x}; \mathbf{s}) = -s_{12}x_1x_3 - s_{23}x_2x_4 - s_{34}x_3x_5 - s_{45}x_1x_4 - s_{15}x_2x_5. \quad (5.29)$$

In the spacelike-collinear limit $\delta \rightarrow 0$, four of the terms remain, with coefficients dependent on each other. We substitute the Mandelstam variables with the parameterization in Eq. (2.2), namely

$$\{s_{12}, s_{34}, s_{45}, s_{15}\} = \{sz, (1-z)xs, s, xs\},$$

obtaining

$$\mathcal{F}(\mathbf{x}; \mathbf{s}) \xrightarrow{\delta=0} \mathcal{F}_0 \equiv (-s)[zx_1x_3 + x(1-z)x_3x_5 + x_1x_4 + xx_2x_5]. \quad (5.30)$$